TECHNICAL DESIGN NOTE 02

Single-model inverse design from HFSS spectra

One neural network | fixed training configuration | no hyperparameter tuning | no inference-time optimization

HFSS CATALOGUE	MODEL INPUT	MODEL OUTPUT
160 standard spectra	**Band + absorption**	**Family + dimensions**

1. Exactly one trainable model

The system receives any band $450 \le a < b \le 1600$ nm and requested minimum absorption $0 \le X \le 1$. It outputs one of four shapes, full-cell dimensions from 0 to 320 nm and an air or SiO2 environment. There is one trainable network, no ensemble, no model comparison, no hyperparameter tuning and no geometry optimization at inference.



The one-network decision

Use a single multi-head residual MLP: one shared feature trunk, one classification output for the 8 geometry/substrate combinations, and one continuous output for the two valid dimensional coordinates. Multiple output heads belong to the same trainable model.

Input request	Output geometry
a: band start (nm)	One of 4 geometry families
b: band end (nm)	Air or SiO2
X: minimum absorption in [0,1]	Outer diameter or side (nm)
Operating condition: fixed TM, 0°	Inner diameter or side (rings only)

No-search specification: one predetermined architecture trained once. A deterministic offline reduction over available validated HFSS spectra generates supervision; there is no optimizer, candidate search or second network at inference. The model output is verified with a separate HFSS solve where possible.

2. Make the HFSS dataset supervise an inverse request

HFSS supplies (geometry, substrate) → absorption spectrum, while the desired ML task is (band, threshold) → geometry. The missing step is deterministic target-label construction from the simulated catalogue.

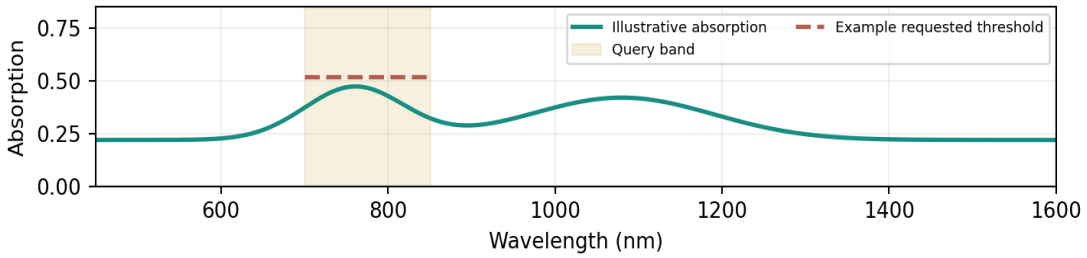


Figure 1. Example showing how a simulated spectrum is assessed against a requested threshold in a wavelength band. Entirely illustrative; not a repository prediction.

$$B(q) = \{\lambda_i : a \le \lambda_i \le b\}$$

$$J_j(q) = \frac{1}{|B|} \sum_{\lambda_i \in B} \max(0, X - A_j(\lambda_i))^2$$

$$j^*(q) = \arg\min_{j \in C(q)} [J_j(q) + \varepsilon P_j]$$

For each synthetic request q, evaluate the stored, validated HFSS catalogue on its 1 nm reporting grid, with interpolated band endpoints where needed. C(q) is the fixed catalogue with any operating constraints. J is zero when every resolved in-band wavelength meets the threshold. P_j is a fixed tie-breaker, such as normalized gold area fraction. The catalogue argmin is an offline label reduction, not an inference-time optimization or an additional learned model. Narrow-band labels inherit the actual HFSS spectral resolution limits.

Training queries

- Generate 10,000 fixed-seed synthetic requests q with continuous 450 ≤ a < b ≤ 1600 nm and 0 ≤ X ≤ 1. Include very narrow and wide bands rather than imposing a 20 nm minimum; flag bands narrower than validated spectral detail as uncertain.
- Include both achievable and unmet requests. Assign the best verified catalogue entry even if its objective J is nonzero; mark feasibility separately.
- Use a balanced request distribution across short, middle and long wavelength intervals, and across demanding/less-demanding thresholds.
- Split geometries before query-label construction to prevent spectra from the same geometry leaking into held-out evaluation.

The 10,000 requests are derived examples from 160 full-wave spectra, not independent full-wave data. Only geometry-partitioned spectra provide genuinely held-out physical designs. For very narrow bands, error depends on the true solved-frequency density, not merely the 1 nm export grid.

3. Output parameterization and fixed architecture

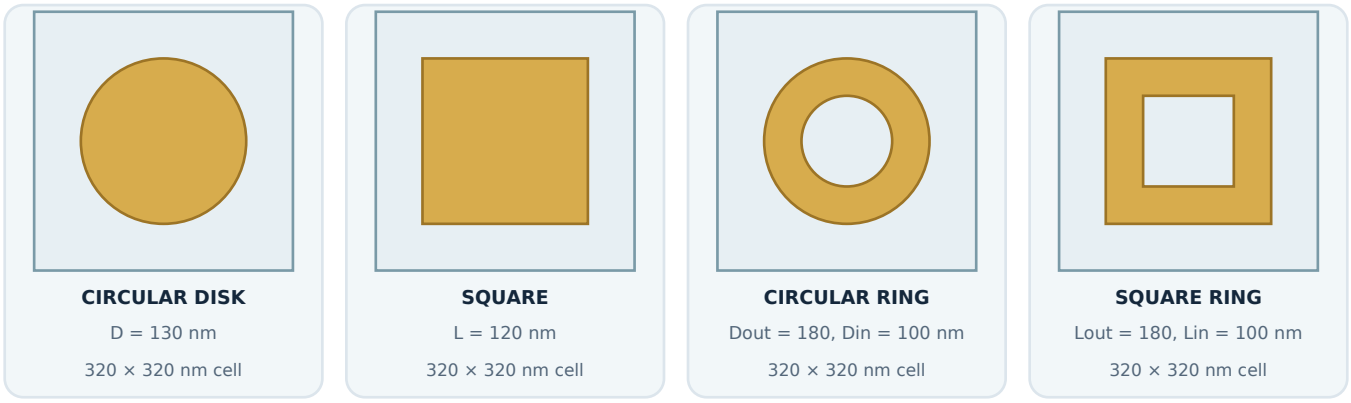


Figure 2. Four candidate geometries, all with unrestricted size within the 320 nm periodic cell. The same mathematical decoder covers disk/ring and square/square-ring families.

Use two normalized coordinate outputs and a deterministic full-domain geometry mapping. Unlike a sigmoid-only decoder, a hard-sigmoid can represent the exact 0 and 1 endpoints, including zero-size and 320 nm contact configurations.

$$d_{\text{out}} = 320 z_o, \qquad d_{\text{in}} = d_{\text{out}} z_i$$

$$z_o = \text{hardsigmoid}(u_o), \quad z_i = \text{hardsigmoid}(u_i)$$

For circles, d is diameter; for squares, d is side length. Solid shapes use d_out = 320 z_o and fix d_in = 0. Rings use both outputs. No geometric rounding or minimum wall/gap is applied. For d_out = 0 or d_in = d_out, the CAD builder must represent zero Au volume without creating a degenerate solid. For d_in = 0, a ring is physically the corresponding solid. The network may return endpoints exactly.

Architecture, selected once

Component	Fixed configuration
Input vector	(a − 450)/1150, (b − 450)/1150, X (3 normalized scalars)
Shared trunk	Linear(3,256) → SiLU → 3 residual blocks (width 256, LayerNorm, SiLU)
Category head	Linear(256,8), softmax over 4 families × 2 environments
Dimensions head	Linear(256,2) → hard-sigmoid → full-domain mapping
Single model	One shared trunk and two heads, trained jointly
Precision	FP32 training; no quantization requirement

Illustrative one-pass model flow



The figure shows the single-pass inference path. The trained network produces a class and valid dimensions without testing candidate geometries at inference.

Numerical limitations at the full boundary

The prediction is allowed to return exact 320 nm and zero-size geometries, as well as extremely narrow positive walls or gaps. For those cases, one forward HFSS verification may be too expensive to converge within the original 14-day data budget. Return the geometry and explicit unverified status rather than clipping it into a narrower domain.

Categorical discontinuity

The same target can be realizable by disk, square or either ring, so the decision boundary between topology classes may be sharp. An 8-way classification head avoids averaging categorical labels, but a single continuous dimension head may still learn ambiguous regressions within the winning category. This remains a genuine limitation of the one-model, no-search constraint.

4. Fixed training protocol and objective

Training must be specified without trying different architectures, model families, loss weights or learning-rate grids. The following is a reproducible starting configuration, not a proven optimum.

Parameter	Prespecified value
Optimizer	AdamW; learning rate 1 × 10^-3; weight decay 1 × 10^-4
Epochs and batch size	200 epochs; batch size 128
Loss	Cross-entropy for 8 classes + Huber for normalized coordinates
Loss weights	1.0 (classification) and 2.0 (dimension); fixed
Random seed	42 for sampling, initialization and training
Training/held-out geometries	64 / 16 geometric settings; all paired substrates stay together
Model selection	None: train once, no HPO, no architecture comparison
Evaluation	Held-out inverse-target error plus HFSS re-simulation of final predictions

$$\mathcal{L} = \mathrm{CE}(\hat{t}, t^*) + 2\,\mathrm{Huber}(\hat{z}, z^*)$$

Feasibility and non-uniqueness

Threshold requests can have several equally good geometric solutions or none at all. Domain endpoints introduce additional equivalent representations: solid = ring with zero inner dimension, and any zero-volume ring is a no-Au limit. Canonicalize physically equivalent labels before training. A fixed catalogue tie-breaker makes labels reproducible but not uniquely physical. Classification and dimension accuracy alone do not prove

that the requested absorption is achieved.

Objective evaluated on physical predictions

For a predicted geometry, report minimum and average simulated absorption over the requested band, and the nonnegative threshold violation. Prefer minimum in-band absorption as the principal physical success metric because it detects narrow absorption gaps hidden by averages.

$$V(q,\hat{g}) = \max\left(0, X - \min_{\lambda_i \in B} A_{\mathrm{HFSS}}(\lambda_i;\hat{g})\right)$$

The reported physical metric must come from an independent HFSS verification solve. Neural classification confidence is not a calibrated probability that the target is physically feasible.

5. Inference, evaluation and limits

At inference, no population, global optimizer, gradient-based geometry search, neural surrogate or second model is used. The one trained MLP provides one geometry directly.

Stage	Operation
1. Request	Receive a, b, X; accept any 450 ≤ a < b ≤ 1600 nm, 0 ≤ X ≤ 1
2. Inference	Normalize (a, b, X); execute exactly one MLP forward pass
3. Decode	Argmax over eight classes; decode hard-sigmoid coordinates, including 0 and 320 nm
4. Return	Report shape, dimensions, substrate, operating assumptions and a feasibility disclaimer
5. Verification	Run HFSS once for the returned geometry when physical correctness matters; verification is not a search

Deliverables and performance measures

- Categorical accuracy: family + substrate, against the deterministic catalogue label.
- Dimensional error: outer/inner geometric parameter error in nm on held-out examples.
- Physical threshold violation V, using full-wave HFSS predictions instead of neural estimates.
- Feasible-request success rate and mean minimum absorption over the requested band.
- Generalization disclosure: normal-incidence TM only; no claims about TE or oblique incidence.

Limits imposed by the chosen restrictions

The baseline contains only 80 geometric parameter settings across the full 0–320 nm domain. Very small gaps, walls and narrow-band resonances are therefore underrepresented or numerically unresolved. Artificially generated query/label pairs do not replace additional electromagnetic diversity. With one direct inverse network and no tuning or runtime search, report observed accuracy and the HFSS verification status instead of asserting that the model finds a globally best geometry.

Two-week dataset connection

The second document uses the same fixed 160 standard HFSS simulations and 8 higher-fidelity checks as Document 01. The 14-day capacity assumption applies only to data generation. Neural training, new held-out verification solves, and potential retries must be scheduled separately unless covered by the reserved verification budget.

References and provenance

[1] Dataset and Problem Formulation, companion PDF, sections 2–6.

[2] Liu et al., Training Deep Neural Networks for the Inverse Design of Nanophotonic Structures. ACS Photonics 2018. https://doi.org/10.1021/acsphotonics.7b01377

[3] Ivan.pdf, Batende.pdf, Yaislin.pdf and repository docs/problem.pdf. GitHub and Ansys source links are in the companion PDF.